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July 20, 1989
SUPERCEDING
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December 1, 1966

FEDERAL SPECIFICATION

BOLTS, (SCREW), LAG

This Federal specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers lag bolts having either square heads or hex heads, for fastening applications.

1.2 Classification.

1.2.1 Types, styles and grades. Lag bolts shall be of the following types, styles, and grades as specified (see 6.2).

Type I - Gimlet point, screw thread.
Type II - Cone point, screw thread.
Type III - Cone point, fetter drive thread.

Style 1 - Hex Head.
Style 2 - Square Head.

Grade A - (Deleted).
Grade B - Commercial steel.
Grade C - Corrosion-resistant steel.
Grade D - Copper-silicon alloy.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Industrial Engineering Group (7FXEI), 819 Taylor St., Fort Worth, TX 76102

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FSC 5306

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Federal Specifications:

QQ-C-591 - Copper - Silicon Alloy, Rod, Wire, Shapes, and Flat Products (Flat Wire, Strips and Bars.)
QQ-P-35 - Passivation Treatment for Austenitic, Ferritic, and Martensitic Corrosion-Resisting Steel (Fastening Devices).
QQ-P-416 - Plating Cadmium (Electrodeposited).
PPP-H-1581 - Hardware (Fasteners and related items), Packaging and Packing for Shipment and Storage of.

Federal Standards:

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.
Fed. Test Method Std. No. 151 - Metals, Test Methods.
Fed. Test Method 151/111.2 - Chemical Analysis.
Fed. Test Method 151/112.2 - Spectrochemical Analysis.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specifications:

MIL-C-13924 - Coating, Oxide, Black, for Ferrous Metals.
DOD-P-16232 - Phosphate Coatings, Heavy Manganese or Zinc Base (For Ferrous Metals).
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No-Go).
MIL-C-81562 - Coating, Cadmium or Zinc (Mechanically Deposited).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS-16992 - Bolt (Screw) Lag, Hex Head, Gimlet Point.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of the invitation for bids or request for proposal shall apply.

ASTM:

ASTM A153 - Zinc Coating (Hot-Dip) On Iron and Steel Hardware.
ASTM A307 - Low-Carbon Steel Externally and Internally Threaded Standard Fasteners.
ASTM A370 - Mechanical Testing of Steel Products.
ASTM B117 - Salt Spray (Fog) Testing.
ASTM A633 - Electrodeposited Coatings of Zinc on Iron and Steel.
ASTM E-18 - Rockwell Hardness of Metallic Materials.

ASTM standards are available from the ASTM, 1916 Race Street, Philadelphia, PA 19103.

American National Standards Institute (ANSI) Standard:

B18.2.1 - Square and Hex Bolts and Screws.

ANSI standards are available from the American National standards Institute, Inc., 1430 Broadway, New York, NY 10018.

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street N.W., Washington, DC 20036).

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W Jackson Blvd., Chicago, IL 60606).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references specified herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Type I, style 1 and style 2. Type I, style 1 and style 2 lag bolts shall conform to the dimensions and tolerances of MS-16922 or ANSI B18.2.1.

3.1.2 Type II, style 1 and style 2. Type II, style 1 and style 2 lag bolts shall conform to the dimensions and tolerances of ANSI B18.2.1.

3.1.3 Type III, style 1 and style 2. Type III, style 1 and style 2 head dimensions and tolerances shall conform to ANSI B18.2.1. The fetter-drive threads shall conform to the dimensions shown in figure 1 and table IV (see 6.4).

3.1.4 Thread dimension tolerances. The dimensional tolerance for those values shown in TABLE 9A of ANSI B18.2.1 shall be plus or minus 0.005 inch.

3.2 Material. Lag bolt material shall be of uniform quality having the mechanical properties specified in 3.5.

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3.2.1 Grade B. When specified (see 6.2), lag bolts shall be low-carbon steel conforming to grade A of ASTM A307.

3.2.2 Grade C. When specified (see 6.2), lag bolts shall be corrosion-resistant steel conforming to the requirements of compositions 302, 303, 304, 305, 310 or 316 specified in Fed. Std. 66.

3.2.2.1 Magnetic permeability. Corrosion-resistant lag bolts shall have a maximum magnetic permeability (m) of 2.0 when tested as specified in 4.5.6.

3.2.3 Grade D. When specified (see 6.2), lag bolts shall be copper-silicon alloy conforming to the requirements of copper alloy numbers 651 or 655 specified in QQ-C-591.

3.3 Sizes.

3.3.1 Nominal diameter and length. Lag bolts shall be of the nominal size (diameter) and lengths as shown in table I, and as specified (see 6.2 and 6.3).

TABLE I Nominal diameter and lengths

** SIZE NO.	DIAMETER IN INCHES (see MS-16992 or ANSI B18.2.1)	** LENGTH NO.	LENGTH IN INCHES
01	NO. 10*	01	1
02	1/4	02	1-1/4
03	5/16	03	1-1/2
04	3/8	04	1-3/4
05	7/16	05	2
06	1/2	06	2-1/2
07	9/16*	07	3
08	5/8	08	3-1/2
09	3/4	09	4
10	7/8	10	4-1/2
11	1*	11	5
12	1-1/8*	12	5-1/2
13	1-1/4*	13	6
		14	6-1/2
		15	7
		16	8
* Applies only to type III lag bolts.		17	9
** Size and length numbers are assigned to facilitate part numbering for cataloging purposes.		18	10
		19	11
		20	12

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3.4 Finish.

3.4.1 Uncoated. Unless otherwise specified (see 6.2), lag bolts shall be furnished uncoated except that a black oxide coating meeting the requirements of MIL-C-13924 will be acceptable.

3.4.2 Zinc coating.

3.4.2.1 Hot-dip, galvanizing. When specified (see 6.2), lag bolts shall be zinc coated by the hot-dip process in accordance with ASTM A153.

3.4.2.2 Electrodeposited zinc. When specified (see 6.2), lag bolts shall be zinc coated by the electroplating process conforming to type II or III, classification Fe/Zn 5 of ASTM B 633.

3.4.2.3 Mechanically deposited zinc. When specified (see 6.2), mechanically deposited zinc coating on lag bolts shall conform to the requirements of type I, class 1, or type II, class 2 of MIL-C-81562.

3.4.3 Cadmium coating. When specified (see 6.2), lag bolts shall be coated with cadmium in accordance with the requirements of type II, class 3, of QQ-P-416.

3.4.4 Phosphate coating. When specified (see 6.2), phosphate coatings on lag bolts shall conform to the requirements of type Z, of DOD-P-16232.

3.4.5 Passivation. When grade C, corrosion-resistant bolts are specified, they shall be passivated in accordance with the requirements of QQ-P-35.

3.5 Mechanical properties. Unless otherwise specified (see 6.2), lag bolts shall have the following mechanical properties when tested as specified in 4.5.2 and 4.5.3.

	Grade B	Grade C	Grade D
Tensile strength, p.s.i. (minimum)	55,000	70,000	55,000
Elongation, in 2 inches,% (minimum)	18	40	10
Hardness Rockwell B (minimum)	60	--	--

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3.6 Salt spray resistance. When specified (see 6.2), shall be tested in accordance with 4.5.5, lag bolts shall show no signs of rust or white salt, except at the crest of threads or at sharp edges.

3.7 Workmanship. Lag bolts shall be uniform in quality and appearance, and shall be free from rust, scale, burrs, fins, seams, and other defects that may affect their serviceability. Screw threads shall be smooth, clean, and true to form. Lag bolts with protective finishes shall be uniform and free from cracks, nicks, blisters or other defects that would affect the quality of the coating.

3.8 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of lag bolts shall be classified as quality conformance inspection.

4.3 Sampling.

4.3.1 Lot. All lag bolts of the same type, style, grade, coating, diameter, and length, offered to the Government at one time, shall be considered a lot for the purpose of inspection.

4.3.2. Sample unit. The sample unit shall be one lag bolt.

4.3.3. Sampling for visual and dimensional examination. A random sample of lag bolts shall be selected from each lot in accordance with MIL-STD-105, at inspection level II. The acceptable quality level shall be 4.0 percent defective.

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4.3.4. Sampling for tests. A random sample of lag bolts shall be selected from each lot in accordance with MIL-STD-105, at special inspection level S-1. The acceptable quality level shall be 2.5 percent defective.

4.3.5. Sampling for chemical composition. When specified (see 6.2), a random sample of lag bolts shall be selected from each lot in accordance with MIL-STD-105, at special inspection level S-1. Failure of any sample to comply with specified requirements shall be cause for rejection of the entire lot.

4.3.6. Sampling for magnetic permeability. When grade C lag bolts are specified (see 6.2), samples shall be selected from each lot in accordance with MIL-STD-105, at special inspection level S-1. The acceptable quality level shall be 2.5 percent defective.

4.3.7. Sampling for coating thickness test. Unless otherwise specified (see 6.2), zinc coated or cadmium coated lag bolts shall be selected from each lot in accordance with MIL-STD-105, at special inspection level S-1. The acceptable quality level shall be 2.5 percent defective.

4.3.8 Sampling for examination of preparation for delivery. A random sample of unit, intermediate, and shipping containers (as applicable) shall be selected from each lot in accordance with MIL-STD-105, at inspection level II. The acceptable quality level shall be 6.5 as defined in terms of defects per hundred unit.

4.4 Examination Each sample selected in accordance with 4.3.3 shall be visually and dimensionally examined to determine conformance with the requirements of this specification (see table II).

TABLE II. Classification of defects

Category	Defects	Reference Paragraph
101	Type, style, or grade not as specified.	1.2.1
102	Diameter not within tolerance.	3.1.1 thru 3.1.3
103	Length not within tolerance.	3.1.1 thru 3.1.3
104	Threads not within tolerance.	3.1.4
105	Protective coating not as specified.	3.4
106	Protective coating not uniform, or free from nicks, or cracks.	3.7
107	Burrs, rust or scale not removed.	3.7

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4.5 Tests.

4.5.1 Chemical analysis test. When specified (see 6.2), samples selected in accordance with 4.3.5 shall be tested by method 111.2 or 112.2 of Fed. Test Method Std. No. 151 to determine if the requirements of 3.2.1, 3.2.2, or 3.2.3 have been met. Failure to meet the requirements shall be cause for rejection of the entire lot.

4.5.2 Tensile strength (wedge-tension method) test. Samples selected in accordance with 4.3.4 shall be tested by the wedge-tension test method specified in ASTM A370 to determine if the requirements of 3.5 have been met. Failure to meet the tensile and elongation requirements shall be cause for rejection of the sample.

4.5.3 Hardness test. When the length of the bolt or the thread is too short for performance of the tension test specified in 4.5.2, the hardness test, specified in ASTM E18 shall be substituted. The hardness of a bolt shall be measured on a transverse section through the threaded portion of the bolt. The hardness shall be determined at a point one-eighth of the nominal diameter from the axis of the bolt and one diameter distance from the end of the bolt.

4.5.4 Thickness of coating. The thickness of the coating on bolts sampled in accordance with 4.3.7, shall be determined in accordance with section 10 of ASTM B 633. Failure to meet the requirements shall be cause for rejection of the sample.

4.5.5 Salt spray test. When specified (see 6.2), sample bolts selected in accordance with 4.3.4 shall be subjected to the salt spray test specified in ASTM B 117. Unless otherwise specified, the test shall last 96 hours. Failure to meet the requirements of 3.6 shall be cause for rejection of the sample.

4.5.6 Magnetic permeability. Samples selected in accordance with 4.3.6 shall be tested with a permeability indicator by the method specified in MIL-I-17214.

4.6 Examination of preparation for delivery. The preservation, packaging, packing, and marking of the lag bolts shall be inspected to determine conformance to the applicable requirements for section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, C or as otherwise specified, (see 6.2).

5.1.1 Level A. The lag bolts shall be preserved and packaged in accordance with the applicable level A requirements of PPP-H-1581.

5.1.2 Level C. The lag bolts shall be preserved and packaged in accordance with the supplier's normal commercial practice, providing that such packaging conforms with rules and regulations applicable to the mode of transportation.

5.2 Packing. Packing shall be level A, B, C, or as otherwise specified, (see 6.2).

5.2.1 Level A. The lag bolts shall be packed in accordance with the applicable level A requirements of PPP-H-1581.

5.2.2 Level B. The lag bolts shall be packed in accordance with the applicable level B requirements of PPP-H-1581.

5.2.3 Level C. The lag bolts shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with the Uniform Freight Classification, and the National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129 as applicable.

6. NOTES

6.1 Intended use. The lag bolts covered by this specification are intended for fastening applications, such as the fastening of metal to wood or wood to masonry.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Part number (see 1.3 and Table I).
(c) Type, style, and grade, of lag bolt required (see 1.2.1).
(d) Size and length of lag bolt required (see 3.3.1 and 6.3).
(e) Kind of finish required, if different (see 3.4).
(f) If different mechanical properties are required (see 3.5).
(g) If chemical composition test is required (see 4.3.5).
(h) If the sampling plan for coated lag bolts is different (see 4.3.7).
(i) If salt spray test is required (see 3.6 and 4.5.5).
(j) Level of preservation, packaging, and marking in section 5 or other requirements (see 5.1 thru 5.3).

6.3 Production sizes. Although lag bolts can be obtained in any size and length when ordered in sufficient quantities, purchases, insofar as practicable, should be confined to the stock sizes and lengths indicated by asterisks in Table III. Galvanized lag bolts normally stocked are indicated by "xx" in Table III.

TABLE III. Stock production sizes of lag bolts (uncoated and galvanized).

Length in inches	Diameter in inches						
	1/4	5/16	3/8	7/16	1/2	5/8	3/4
1	(*)	(*)	(*)	--	--	--	--
1-1/4	(*)	(*)	(*)	--	--	--	--
1-1/2	(*) xx	(*) xx	(*) xx	(*)	(*)	--	--
1-3/4	(*)	(*)	(*)	--	--	--	--
2	(*) xx	(*) xx	(*) xx	(*)	(*) xx	(*)	--
2-1/2	(*) xx	(*) xx	(*) xx	(*)	(*) xx	(*)	(*)
3	(*) xx	(*) xx	(*) xx	(*)	(*) xx	(*)	(*)
3-1/2	(*)	(*)	(*)	(*)	(*)	(*)	(*)

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TABLE III. (CONTINUED)

Length in inches	Diameter in inches 1/4	5/16	3/8	7/16	1/2	5/8	3/4
4	(*)	(*)	(*)	(*)	xx (*)	(*)	(*)
4-1/2	(*)	(*)	(*)	(*)	xx (*)	(*)	(*)
5	(*)	(*)	(*)	(*)	(*)	(*)	(*)
5-1/2	--	(*)	(*)	--	(*)	(*)	--
6	(*)	(*)	(*)	(*)	(*)	(*)	(*)
6-1/2	--	--	--	--	(*)	--	--
7	--	--	(*)	--	(*)	(*)	(*)
8	--	--	(*)	--	(*)	(*)	(*)
9	--	--	--	--	(*)	(*)	(*)
10	--	--	--	--	(*)	(*)	(*)
11	--	--	--	--	--	(*)	(*)
12	--	--	--	--	--	(*)	(*)

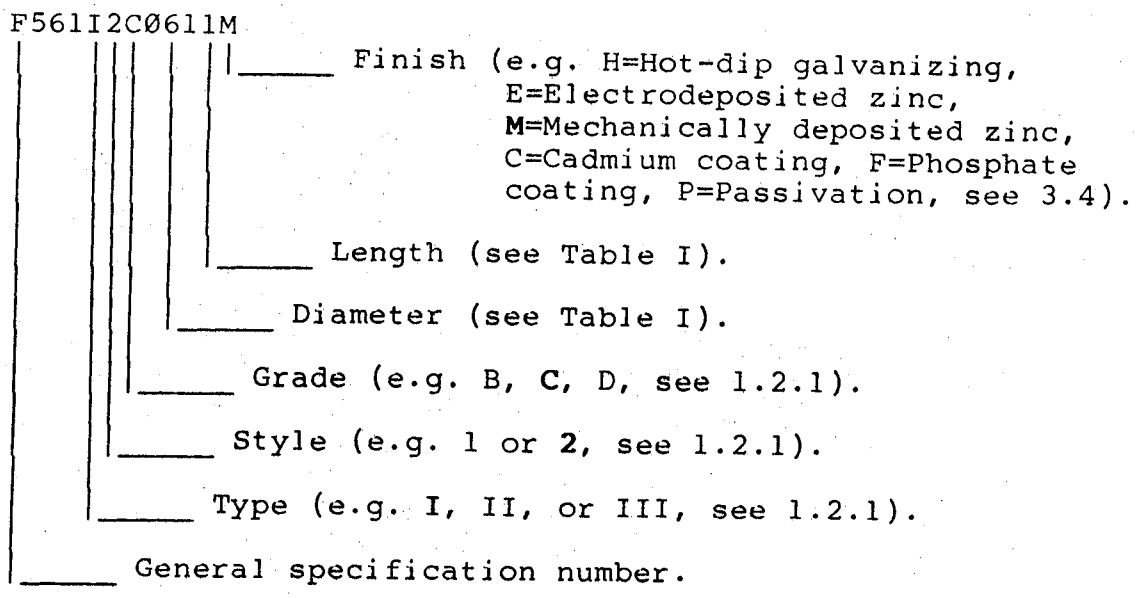
* Uncoated lag bolts.
xx Stock production sizes of galvanized bolts.

6.4 Fetter drive threads. No generally recognized standard for fetter drive threads exists which accounts for variance in the number of threads per inch and thread form. Figure 1 presents a recommended thread design and dimensions which may ultimately provide the basis of a standard. The only limits recommended in figure 1 are the minimum minor diameter and the maximum crest width. The root radius is not critical but should preferably be not less than basic.

6.5 Part Number. Lag bolts conforming with this specification shall be identified by a part number configuration consisting of identification of the general specification number, type, style, grade, diameter, length, and finish (see next page). This part numbering system is intended for identification and cross-indexing of the item within the Federal cataloging system. Part numbers are not required to be placed on the product or container.

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EXAMPLE:



MILITARY INTERESTS:

Military custodian:

Air Force - 99

Review activities:

Air Force - 82
Navy - MC

User activities:

Army - AR

PREPARING ACTIVITY:

GSA - FSS (7FXE)

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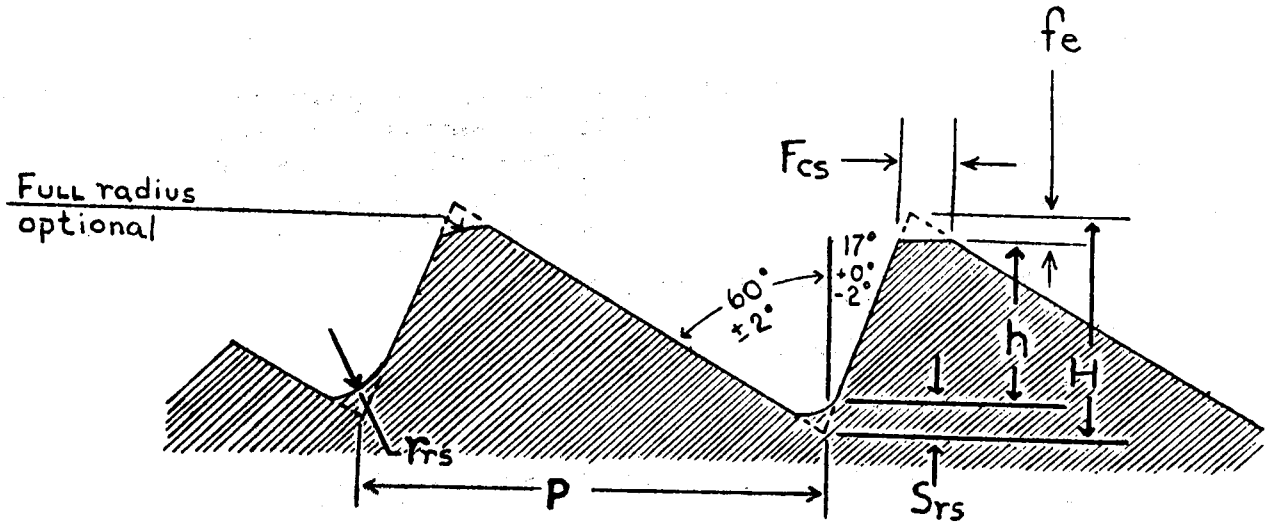


Figure 1. Recommended Thread Form for Fettered Drive Threads (see table IV).

TABLE IV. Recommended Fetter Drive Thread dimensions (see fig. 1)

Nominal size	Threads per inch, n	Pitch P	Depth of fundamental triangle H=0.49073p	Depth of thread, basic, h=0.3750p	Major diameter, basic D	Minor diameter, min., K= D-2h	Crest & root truncation, basic, Fcs= Srs=0.05786p	Width of flat at crest		Root radius	
								Basic Fcs= 0.11792p	Max. Fcs= 0.125p	Basic rrs= 0.1170p	Approx.
1	2	3	4	5	6	7	8	9	10	11	12
		Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch
No.10	13	.07692	.03775	.02885	.1900	.1323	.0045	.0091	.0096	.0090	.010
1/4	11	.09091	.04461	.03409	.2500	.1818	.0053	.0107	.0114	.0106	.010
5/16	9	.11111	.05453	.04167	.3125	.2292	.0064	.0131	.0139	.0130	.015
3/8	8	.12500	.06134	.04688	.3750	.2812	.0072	.0147	.0156	.0146	.015
7/16	7	.14286	.07010	.05357	.4375	.3304	.0083	.0168	.0179	.0167	.015
1/2	6	.16667	.08179	.06250	.5000	.3750	.0096	.0197	.0208	.0195	.020
9/16	6	.16667	.08179	.06250	.5625	.4375	.0096	.0197	.0208	.0195	.020
5/8	5	.20000	.09815	.07500	.6250	.4750	.0116	.0236	.0250	.0234	.025
3/4	4-1/2	.22222	.10905	.08333	.7500	.5833	.0129	.0262	.0278	.0260	.025
7/8	4	.25000	.12268	.09375	.8750	.6875	.0145	.0295	.0312	.0292	.030
1	3-1/2	.28571	.14021	.10714	1.0000	.7857	.0165	.0337	.0357	.0334	.036
1-1/8	3-1/4	.30769	.05099	.11538	1.1250	.8942	.0178	.0363	.0385	.0360	.036
1-1/4	3-1/4	.30769	.05099	.11538	1.2500	1.0192	.0178	.0363	.0385	.0360	.036

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